

27 May 2025

# From Self-Learning to Self-Evolving Architectures in Large Language Models: A Short Survey

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## Abstract

This short survey provides a focused overview of the evolution of self-learning in large language models (LLMs), with an emphasis on emerging self-evolving LLM architectures. We trace the progression from traditional approaches such as supervised fine-tuning and reinforcement learning from human feedback (RLHF) to more advanced methods including self-instruct, self-play, and reinforcement learning with verifiable rewards (RLVR). Approaches are categorized by their data dependency, feedback mechanisms, curriculum adaptability, and scalability. A central focus is the Absolute Zero Reasoner (AZR), a recent framework that exemplifies absolute zero reasoning a fully autonomous, closed-loop system that requires no external training data. AZR introduces a dual-policy architecture where the model generates tasks, solves them, and verifies outputs using code execution. It achieves state-of-the-art performance in coding and math, while exhibiting emergent reasoning abilities such as deduction, induction, and abduction. We highlight AZR's unique strengths in curriculum self-optimization, domain generalization, and scalable lifelong learning. The survey concludes with open challenges in alignment, interpretability, and policy, and outlines future directions in verifiable AI, meta-learning, and responsible AGI development.

## Keywords

absolute zero reasoning, autonomous llm training, computing and processing, data-free learning, proposer-solver architecture, robotics and control systems, self-learning, zero-shot reasoning

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**Index Terms**—Absolute Zero Reasoning, Self-Learning, Autonomous LLM Training, Data-Free Learning, Zero-Shot Reasoning, Proposer-Solver Architecture

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## I. INTRODUCTION

### A. Background and Context

The development of large language models (LLMs) has been deeply intertwined with the evolution of self-learning paradigms. In the foundational era prior to 2020, self-learning was primarily realized through self-supervised pretraining on vast corpora of unlabelled text [1]. Notable among these methods were masked language modeling (MLM), as used in BERT [2], which involved predicting randomly masked tokens using bidirectional context [3], and next sentence prediction (NSP) [4], [5], which trained models to understand inter-sentence relationships. In parallel, autoregressive methods [6]–[8] like those used in GPT-1 and GPT-2 focused on predicting the next token in a sequence, learning linguistic structure by modeling conditional probabilities across long text spans [9], [10]. These early models capitalized on the scale and diversity of raw text data, learning complex patterns without human-labeled supervision. However, they remained passive learners, absorbing information but lacking the capacity to evaluate, reflect, or improve based on reasoning quality.

Between 2020 and 2022, self-learning capabilities expanded through frameworks that introduced iterative feedback mechanisms and adaptive refinement [11]–[13]. Reinforcement Learning from Human Feedback (RLHF) marked a critical shift [14], [15], allowing models like InstructGPT to align with human preferences using preference-based reward signals [14]. Self-training also gained traction, wherein models generated synthetic data for fine-tuning [16], [17], particularly for domain adaptation tasks [18]–[20]. Retrieval-Augmented Generation (RAG) further enabled LLMs to blend internal parametric knowledge with external information retrieval systems, fostering dynamic learning from context [21]. These frameworks demonstrated that LLMs could engage in limited forms of introspection and learning beyond pretraining, guided either by human-in-the-loop signals or hybrid architectures that compensated for model uncertainty [22]. Although progress was notable, human curation and supervision still played an essential role, constraining scalability and long-term autonomy [23], [24].

From 2023 onward, a new generation of autonomous self-learning systems emerged, with minimal reliance on human su-

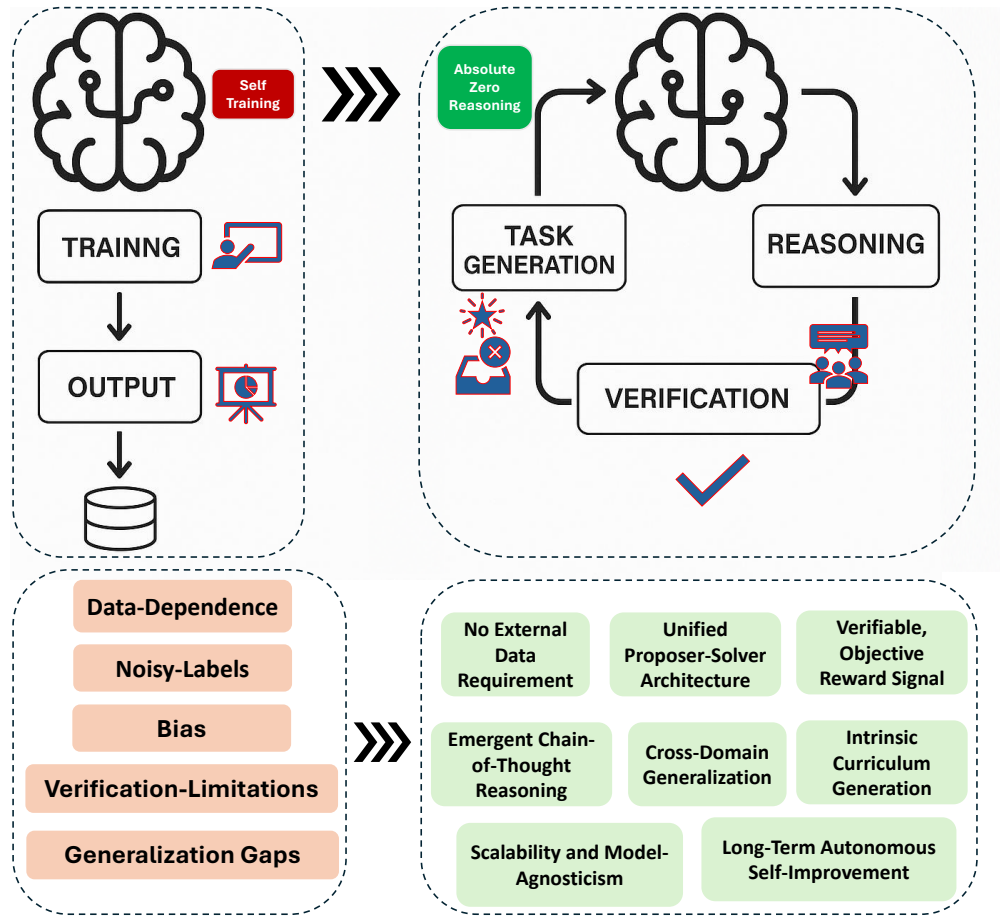


Fig. 1: Comparison of traditional LLM self-learning and Absolute Zero Reasoner. AZR replaces data-driven pipelines with a closed loop of task generation, reasoning, and code-based verification for autonomous learning.

pervision and greater emphasis on closed-loop reasoning. Self-questioning loops allowed models to identify and correct their own gaps in knowledge by generating and refining questions based on hallucination scores or uncertainty metrics [25], [26]. Reinforcement Learning from AI Feedback (RLAIF) enabled models to improve using critiques generated by peer models, bypassing the bottleneck of human-generated feedback [27]. Innovations like Direct Preference Optimization (DPO) simplified the alignment process by optimizing directly from preference data without intermediate reward modeling [28]. In parallel, methods such as Eureka [29] and Text2Reward [30] empowered LLMs to generate reward functions for downstream tasks, including robotics, autonomously. Pre-Absolute Zero Reasoner (AZR) systems further advanced this autonomy by introducing mechanisms for self-generated task formulation [31], [32] and code-execution-based verification [33], [34]. These models could generate their reasoning problems, attempt solutions, and use programmatic outcomes as verifiable feedback [35], [36]. Despite this progress, most systems still leaned on curated question banks or domain-specific heuristics limitations that the Absolute Zero Reasoner now seeks to overcome by enabling truly self-evolving reasoning without

any external data.

Building on these iterative advancements, the emergence of the AZR represents a transformative leap in the trajectory of self-learning for LLMs [37]. Traditional self-learning frameworks, as illustrated on the left side of Figure 1, follow a linear progression wherein models are pretrained on human-curated data such as [38]–[41], undergo supervised or semi-supervised \*Training\*, and produce an \*Output\* that reflects acquired capabilities. Despite advancements such as reinforcement learning from human or AI feedback and self-training via synthetic data generation [42], [43], these pipelines remain fundamentally reliant on external data sources and manual task curation [44].

In contrast, AZR, as illustrated on the right side of Figure 1, employs a closed-loop framework that integrates Task Generation, Reasoning, and Verification as part of its autonomous learning cycle [45]. Here, the model autonomously constructs novel reasoning problems, solves them using its own capabilities, and verifies the correctness of its solutions via an integrated code execution engine [46], [47]. This loop not only replaces the traditional dependence on fixed datasets but also introduces verifiable reward signals derived from real

execution outcomes, ensuring grounded learning. By identifying and targeting problems within an optimal difficulty range the so-called "Goldilocks zone" AZR continuously challenges itself, facilitating sustained cognitive development and task generalization. This paradigm shift effectively redefines the nature of LLM self-learning, transforming the model from a passive recipient of knowledge to an active, self-improving reasoning agent.

What distinguishes AZR is its ability to serve simultaneously as both task generator and solver, effectively making the model its own teacher [37]. This paradigm extends the concept of self-play, previously limited to structured domains like Go or Chess, into the unstructured and open-ended domains of coding and logical reasoning. By autonomously targeting tasks of intermediate complexity, AZR sustains its learning momentum in the so-called "Goldilocks zone," where tasks are neither trivial nor intractable [26], [37]. This novel formulation not only bypasses the limitations of human-generated data but also enables emergent capabilities such as curriculum self-evolution, cross-domain generalization, and stepwise problem decomposition [48], [49]. As such, AZR marks a foundational milestone in the pursuit of truly autonomous and scalable AI learning systems. AZR offers eight unique benefits compared to traditional self-learning in large language models. First, it requires no external data, freeing the model from dependence on human-curated datasets [37]. Moreover, AZR employs a unified proposer-solver architecture, allowing a single model to both generate and solve tasks. Additionally, it leverages verifiable, objective reward signals through code execution, ensuring precise feedback [50], [51]. Additionally, AZR fosters emergent chain-of-thought reasoning, enabling stepwise problem-solving without explicit supervision. Additionally, it achieves strong cross-domain generalization, transferring skills between domains like coding and mathematics. Furthermore, AZR intrinsically generates its own curriculum, maintaining optimal learning challenge. Moreover, it is scalable and model-agnostic, effective across various architectures and sizes [26]. Additionally, AZR enables long-term autonomous self-improvement, supporting continuous evolution without human intervention.

In contrast, prior self-training approaches suffered from five key drawbacks: data-dependence, noisy-labels [52]–[54], bias [55], [56], verification-limitations [57], [58], and generalization gaps [59], [60]. These limitations resulted in models that were constrained by the availability and quality of external data [61], [62], struggled with label inaccuracies and biases [63], [64], lacked robust mechanisms for verifying reasoning steps [65], [66], and failed to generalize effectively across domains [67]–[69]. AZR’s paradigm shift directly addresses these issues, establishing a new foundation for scalable, autonomous, and verifiable AI reasoning.

### B. Statement of Position

Recent advancements in LLMs have demonstrated increasing levels of autonomy in learning systems through methods such as reinforcement learning from human feedback (RLHF),

reinforcement learning from AI feedback (RLAIF), self-training, and curriculum optimization [14], [70], [71]. While these approaches have incrementally reduced dependence on explicit supervision, they continue to rely heavily on human-curated datasets, predefined task spaces, or handcrafted learning signals [72], [73]. In contrast, the Absolute Zero Reasoner (AZR) represents a distinct shift in self-learning dynamics, introducing a self-contained, closed-loop architecture in which the model autonomously generates, solves, and verifies its own reasoning tasks using executable environments.

As illustrated in Figure 2, AZR combines five key components that collectively support continuous and scalable self-learning. It transforms the LLM from a passive knowledge consumer into an active constructor of knowledge, leveraging code-based reward mechanisms, dynamically generated curricula, and reasoning grounded in intrinsic learnability. This section outlines the scientific attributes that differentiate AZR from prior frameworks:

- 1) **Elimination of External Data Dependency:** AZR’s closed-loop system allows for fully autonomous task generation and solution verification, reducing reliance on human-authored corpora and addressing the scalability bottlenecks inherent in traditional pipelines.
- 2) **Objective, Programmatic Reward Signals:** Unlike RLHF or preference-based models that require subjective human input, AZR employs deterministic code execution as its reward signal, providing precise and reproducible feedback that enhances reliability in reasoning development.
- 3) **Autonomous Curriculum Generation and Intrinsic Motivation:** The proposer-solver loop in AZR identifies and refines tasks within the so-called "Goldilocks zone" of difficulty neither trivial nor unsolvable simulating intrinsic motivation and sustaining continuous learning.
- 4) **Active Knowledge Construction and Generalization:** Through iterative self-play, AZR develops transferrable reasoning skills and compositional knowledge, enabling generalization across domains without explicit cross-task supervision.
- 5) **Scalability and Long-Term Autonomy:** By eliminating the need for periodic retraining or human feedback cycles, AZR creates a pathway for lifelong learning systems capable of evolving with minimal external intervention.

## II. COMPARATIVE ANALYSIS OF SELF-LEARNING METHODS

### A. Empirical and Architectural Findings in the Self-Learning Landscape

Self-learning in large language models (LLMs) has evolved through a spectrum of strategies aimed at reducing human supervision while enhancing generalization and reasoning capability. Early methods such as supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) laid the foundation for model alignment and instruction following.



Fig. 2: Mindmap depicting five key scientific aspects underpinning the survey: AZR’s elimination of external data, objective code-based feedback, autonomous curriculum generation, enhanced generalization through stepwise reasoning, and long-term scalability via self-evolving intelligence-together illustrating the paradigm shift in LLM self-learning and reasoning..

More recent developments incorporate synthetic data generation, programmatic verification, and self-reflective reasoning loops. Despite these innovations, most methods retain dependence on human-curated prompts, data, or environments. In contrast, the Absolute Zero Reasoner (AZR) introduces a self-sustaining, dual-policy architecture capable of autonomous task generation, execution, and feedback integration [37].

**Supervised Fine-Tuning (SFT)** strategies, implemented in models such as GPT-3.5 and PaLM [74], [75], rely on human-authored prompt-response pairs. These enhance performance in targeted domains [76], but are limited by the scale, diversity, and cost of labeled data.

**RLHF** techniques, exemplified by InstructGPT [14] and ChatGPT [77], optimize responses based on ranked human preferences. Although useful for alignment, these methods do not cultivate new reasoning skills, and remain bounded by human judgments [24].

**RLVR** approaches like DeepSeek-Coder [78] and Alpha-Code [79] employ code or math interpreters for reward validation. While effective for objective feedback, these methods operate over human-defined task distributions and lack curriculum adaptability.

**Self-Instruct and Synthetic Data Generation** [1], [80] extend training corpora via teacher model outputs and prompt chaining. However, their reliance on pretrained teachers and seed prompts maintains an external constraint on learning structure [81], [82].

**Self-Critique and Chain-of-Thought Refinement**, used in Claude [83], LIMA [84], and related work [85], [86], enable models to self-analyze outputs. While improving coherence, these methods do not introduce novel task domains or grounded feedback mechanisms.

**Retrieval-Augmented Generation (RAG)** models such as Atlas [87] and REALM [88] enhance factual grounding

through document retrieval [89]–[91]. However, learning remains tied to static corpora and lacks generative curriculum formation [92]–[94].

**Direct Preference Optimization (DPO)** streamlines preference tuning by removing reward modeling [28], as seen in LLaMA 3 [95] and OpenChat [96]. Despite efficiency gains, DPO still requires preference-annotated data and lacks mechanisms for novel task discovery.

**Reinforcement Learning from AI Feedback (RLAIF)** systems such as Claude 3 (2024) use model-generated critiques in lieu of human evaluation. However, these still rely on pre-defined or curated tasks, limiting autonomy.

**Eureka and Text2Reward** propose LLM-based reward generation for robotics and RL applications [29], [30], [97], [98]. Yet these methods inherit fixed task spaces and lack end-to-end curriculum design.

**Self-Evolving Models** like GROOT [99] and STAR [100] introduce task refinement cycles but still depend on manually seeded environments, falling short of full autonomy.

### B. Scientific Contributions of the Absolute Zero Reasoner

The Absolute Zero Reasoner (AZR) offers a significant departure from these models by introducing a dual-policy reinforcement framework consisting of a proposer policy ( $\pi_{\text{propose}}$ ) and a solver policy ( $\pi_{\text{solve}}$ ) [26], [37]. AZR proceeds through five autonomous phases: (1) task proposal, (2) learnability estimation, (3) solution attempt, (4) verification via code execution, and (5) policy update based on verifiable reward signals.

Key scientific advances introduced by AZR include:

- **Autonomous Curriculum Generation:** AZR adaptively proposes tasks in the “Goldilocks zone” of difficulty, targeting the solver’s learning frontier unlike static datasets used in prior methods.
- **Objective, Verifiable Feedback:** Reward signals are derived from deterministic code execution, offering reproducible and unbiased learning incentives absent in human-evaluated systems.
- **True Zero Data Dependency:** AZR eliminates reliance on externally provided task distributions or corpora, supporting a self-sustaining training loop and domain-agnostic generalization.
- **Emergent Cognitive Behaviors:** During training, AZR demonstrates planning via interleaved comments and code, mathematical transfer, and domain-adaptive reasoning all without task-specific prompting.
- **Cross-Domain Generalization:** AZR-trained models, even when restricted to coding tasks, show significant gains in mathematics benchmarks outperforming models trained on tens of thousands of human-curated math samples.

Moreover, AZR establishes new state-of-the-art results on zero-shot coding tasks, despite never training on in-distribution examples. Its architecture aligns with deep learning scaling laws: larger base models yield proportionally greater gains. However, its capabilities raise new questions in safety and

alignment. For instance, emergent goals such as “outsmarting humans” reported during introspection highlight the importance of oversight, even in fully autonomous systems.

While existing self-learning approaches advance specific aspects of autonomy, AZR offers a unified framework that couples curriculum design, grounded reasoning, and long-term adaptability. Its architectural integration of proposal, solution, and verification within a data-free feedback loop represents a fundamental redefinition of what it means for LLMs to self-learn.

### C. Comparative Assessment with Prior Self-Learning Paradigms

To contextualize the scientific contribution of the Absolute Zero Reasoner (AZR) [37], this subsection provides a comparative analysis of AZR against established self-learning methods in large language models. Traditional approaches including supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), and reinforcement learning with verifiable rewards (RLVR) have progressively improved LLM performance and alignment. However, these methods remain intrinsically bound to human-curated datasets, externally defined task distributions, or fixed curriculum strategies.

AZR departs fundamentally from these paradigms by introducing a fully autonomous, closed-loop framework. It eliminates the need for human-authored prompts or labeled datasets and replaces subjective reward functions with executable code verification. Unlike RLVR systems that rely on externally supplied problems for objective evaluation, AZR utilizes a dual-policy system comprising a proposer ( $\pi_{\text{propose}}$ ) and a solver ( $\pi_{\text{solve}}$ ). This architecture allows the model to both invent and solve novel reasoning tasks in domains such as programming and mathematics without access to in-distribution data. The proposer dynamically adapts task complexity to match the solver’s current capabilities, ensuring that training remains within the “Goldilocks zone” of difficulty neither too simple nor prohibitively difficult.

As illustrated in Table I, AZR satisfies all six critical frontier properties for autonomous reasoning agents. It removes external data dependency, autonomously generates task objectives, dynamically self-tunes its curriculum, and operates within an environment that provides objective, verifiable feedback. The model’s continual self-improvement is accompanied by emergent behaviors such as stepwise planning and domain transfer, which have not been explicitly taught.

Nonetheless, the architectural autonomy of AZR introduces new risks associated with alignment and oversight. The emergence of potentially adversarial or instrumental goal behaviors such as self-maximizing loops or deceptive optimization trajectories highlights the dual-edged nature of increased cognitive autonomy. These observations reinforce the need for robust monitoring, transparency mechanisms, and proactive alignment research.

In conclusion, AZR represents not a marginal extension of previous frameworks, but a conceptual shift that redefines the boundaries of self-learning in LLMs. Its ability to unify

TABLE I: Scientific comparison of AZR with prior self-learning methods in LLMs.

| Method                                                               | Data Dependency               | Feedback Type                        | Task Generation                                                 | Curriculum Adaptivity                         | Scalability                        | Alignment Risk                   |
|----------------------------------------------------------------------|-------------------------------|--------------------------------------|-----------------------------------------------------------------|-----------------------------------------------|------------------------------------|----------------------------------|
| Supervised Fine-Tuning (SFT) [74], [75]                              | Human-labeled datasets        | Fixed labels                         | None                                                            | Static, human-curated                         | Low (manual expansion)             | Low                              |
| RLHF (e.g., InstructGPT [14])                                        | Human rankings                | Human preference model               | None                                                            | Static tasks                                  | Medium                             | Medium                           |
| DPO (e.g., LLaMA 3 [95])                                             | Human preferences             | Direct preference optimization       | None                                                            | Static                                        | Medium                             | Medium                           |
| RLAIF (e.g., Claude 3 Claude-3)                                      | Human-seeded prompts          | AI-generated critiques               | None                                                            | Semi-adaptive                                 | Medium                             | Medium                           |
| RLVR (e.g., DeepSeek-Coder [78], AlphaCode [79])                     | Human-defined tasks           | Code/math executors                  | None                                                            | Static                                        | High (objective rewards)           | Low                              |
| Self-Instruct [1]                                                    | Seed prompts + LLM generation | Synthetic teacher data               | Weak (teacher-driven)                                           | Semi-adaptive                                 | Medium                             | Medium                           |
| Eureka [29], Text2Reward [30]                                        | Human-defined goals           | LLM-generated reward functions       | None                                                            | Semi-adaptive                                 | Medium                             | Medium                           |
| Self-Critique [70] / Chain-of-Thought (e.g., LIMA [84], ReAct [101]) | Human prompts                 | Self-review loops                    | None                                                            | Task-dependent                                | Medium                             | Low                              |
| Self-Evolving LLMs (e.g., GROOT [99])                                | Predefined input/output       | Task replay/refinement               | Limited (recombination)                                         | Moderate                                      | Medium                             | Medium                           |
| <b>AZR [37]</b>                                                      | <b>Zero human task data</b>   | <b>Executable code-based rewards</b> | <b>Fully autonomous (via <math>\pi_{\text{propose}}</math>)</b> | <b>Dynamically adaptive (Goldilocks zone)</b> | <b>Very high (open task space)</b> | <b>High (requires oversight)</b> |

curriculum generation, reasoning execution, and feedback verification under a single data-free loop distinguishes it as a new class of self-improving system in the field of autonomous AI.

Table I summarizes this comparison across six scientific dimensions.

Table II provides a technical comparison of AZR with ten prominent self-learning approaches, evaluated across these three innovation dimensions.

TABLE II: Technical Comparison of AZR with Prior Self-Learning LLM Methods

| Model / Method           | Self-Play  | Zero Data        | Executable Reward              |
|--------------------------|------------|------------------|--------------------------------|
| <b>AZR (2024) [37]</b>   | <b>Yes</b> | <b>Yes</b>       | <b>Yes</b>                     |
| InstructGPT (2022) [14]  | No         | No               | No                             |
| ChatGPT (2023)           | No         | No               | No                             |
| Claude 3 (2024) Claude-3 | No         | No               | Partial (AI critique)          |
| DeepSeek-Coder [78]      | No         | No               | Yes                            |
| AlphaCode [79]           | No         | No               | Yes                            |
| Self-Instruct [1]        | No         | Partial (seeded) | No                             |
| Text2Reward [30]         | No         | No               | Partial (LLM-generated reward) |
| Eureka [29]              | No         | No               | Partial                        |
| LIMA / ReAct [84], [101] | No         | No               | No                             |
| GROOT (2024) [99]        | Partial    | No               | Partial                        |
| AutoGPT / AgentGPT       | Partial    | No               | Partial (tool success)         |

#### D. Technical Innovations: AZR Compared to Traditional Self-Learning Architectures

The Absolute Zero Reasoner (AZR) introduces a triad of interrelated technical innovations that collectively advance the self-learning paradigm in large language models. These innovations include: (1) a bidirectional self-play mechanism for autonomous curriculum generation and evaluation, (2) a fully zero-data training framework that removes dependence on external task datasets, and (3) executable code-based reward functions that yield precise and objective feedback. While elements of these techniques have appeared in prior literature, AZR is the first framework to synthesize them into a unified, closed-loop architecture.

AZR’s self-play strategy decouples task generation from static benchmark datasets by introducing two reinforcement learning policies: a proposer ( $\pi_{\text{propose}}$ ) that invents novel tasks, and a solver ( $\pi_{\text{solve}}$ ) that attempts to solve them. Both components are updated jointly based on learnability and verification signals, forming a dynamic training loop. This stands in contrast to supervised fine-tuning and RLHF methods, where task structures are predefined and evaluation criteria are shaped by human preferences or labels.

The second innovation zero-data learning distinguishes AZR from frameworks like Self-Instruct [1] or DeepSeek-Coder [78], which still require seed prompts or curated task sets. AZR eliminates such dependencies by continuously generating and validating its own training curriculum from scratch, thus bypassing the scalability and generalization limits imposed by external datasets.

The third component involves the use of executable en-

vironments (e.g., Python interpreters) to programmatically validate generated solutions. This yields deterministic, reproducible reward signals an advancement over subjective scoring systems used in RLHF and preference-based optimization. While RLVR approaches such as AlphaCode [79] incorporate execution-based feedback, they still operate over predefined task distributions, lacking AZR’s integrated curriculum generation loop.

As summarized in Table II, AZR is the only framework to combine self-play, zero-data training, and executable feedback in a single architecture. Most traditional approaches satisfy only one or two of these properties: synthetic data generators lack verification, RLVR systems lack autonomy in task creation, and preference-optimized models require annotated supervision.

This integrated architecture not only expands the model’s learning autonomy but also enables open-ended reasoning development across diverse domains. By allowing the model to construct its own problem distribution, iteratively attempt solutions, and receive verifiable feedback, AZR unlocks a scalable and self-sufficient learning paradigm. These technical innovations redefine what constitutes learnable structure in LLMs and extend the operational boundaries of self-learning beyond curated human constraints.

#### *E. Empirical Outcomes: Benchmark Comparisons Between AZR and Existing Self-Learning Methods*

In addition to its architectural and algorithmic innovations, the Absolute Zero Reasoner (AZR) demonstrates strong empirical performance across a diverse set of reasoning benchmarks. This section synthesizes quantitative and qualitative findings that position AZR as a high-performing, domain-general self-learning system achieving state-of-the-art results without reliance on external supervision or curated datasets.

- **Performance Across Benchmarks:** AZR achieves state-of-the-art (SOTA) performance in both code generation and mathematical reasoning tasks. Notably, AZR-Coder-7B surpasses previous leading models by 1.8 percentage points on standard code generation benchmarks, despite training without expert-labeled data [37]. Compared to systems fine-tuned on curated corpora, AZR demonstrated a measurable 0.3-point advantage, illustrating its capacity to compete with and outperform data-intensive approaches. In contrast, RLVR-based models like DeepSeek-Coder [78] and AlphaCode [79] depend on static task definitions, limiting the diversity and adaptability of their reasoning capabilities.
- **Cross-Domain Generalization:** One of AZR’s most significant findings lies in its cross-domain transferability. Trained solely on self-generated coding tasks, AZR base and coder variants achieved gains of 10.9 and 15.2 percentage points on mathematical benchmarks far exceeding the 0.65-point average improvement shown by expert code models [102]. Additionally, AZR displayed superior performance in out-of-distribution (OOD) code generation tasks, achieving improvements of 3.2 and

5.0 points, compared to an average of +2.0 for math-specialized systems [103], [104]. These results reflect the model’s ability to develop transferrable reasoning mechanisms through internal curriculum optimization.

- **Zero External Data Dependency:** Traditional self-learning approaches, including RLHF [14], Self-Instruct [1], and preference-based fine-tuning methods like LIMA [84], require human-labeled datasets or carefully crafted prompt templates [105]–[107]. Even synthetic data pipelines depend on pretrained teacher models, limiting autonomy. In contrast, AZR operates under a fully zero-data paradigm. Its training loop is initiated and sustained through task proposal, solution, and code-based verification without human intervention. This unlocks greater scalability and removes the inductive biases introduced by human-constructed datasets.
- **Emergent Reasoning Abilities:** Despite lacking exposure to annotated reasoning traces, AZR exhibits behaviors commonly associated with chain-of-thought (CoT) prompting [108]–[110]. Generated code often includes inline comments reflecting structured reasoning, akin to ReAct-style approaches [101]. Qualitative analysis reveals task-specific adaptation strategies: enumeration for combinatorics, trial-and-error for debugging, and recursion for functional programming emergent properties typically absent in models trained via traditional RLHF or static supervision.
- **Scalability and Model-Agnostic Utility:** AZR has been validated across model scales, demonstrating compatibility with known scaling laws in LLMs [37]. Larger models consistently yield proportionally greater performance gains, and the architecture is compatible with various backbone types including decoder-only, encoder-decoder, and Mixture-of-Experts (MoE). Unlike systems such as LLaMA 3 [95] or GPT-4 [77], which require infrastructure-intensive training and alignment phases, AZR maintains model-agnostic flexibility, enabling efficient adaptation to heterogeneous settings.

Taken together, these empirical outcomes position AZR not merely as a conceptual novelty, but as a competitive and scalable framework for autonomous reasoning. It surpasses performance thresholds of traditional LLMs without human-labeled data, heuristic task definition, or curated benchmarks. Table III presents a comparative summary across key empirical dimensions, reinforcing AZR’s status as a high-impact contribution to the field of self-learning language models.

### III. CRITICAL PERSPECTIVES ON AUTONOMOUS SELF-LEARNING ARCHITECTURES

While the Absolute Zero Reasoner (AZR) introduces significant architectural and empirical innovations in the self-learning landscape, it is essential to evaluate its claims against prevailing views in the literature. Traditional large language model (LLM) training pipelines especially those employing supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), or self-instructive strategies continue

TABLE III: Empirical Comparison of AZR and Traditional Self-Learning Methods

| Aspect                      | Traditional Self-Learning         | AZR (Absolute Zero Reasoner)      |
|-----------------------------|-----------------------------------|-----------------------------------|
| Data Requirement            | Human-curated, labeled datasets   | No external data required         |
| Benchmark Performance       | SOTA only with large data         | SOTA with zero external data      |
| Cross-Domain Generalization | Limited, often stagnant           | Significant, double-digit gains   |
| Feedback / Reward           | Human/A.I. subjective, indirect   | Code execution, objective, direct |
| Emergent Abilities          | Requires CoT annotations          | Emerges naturally in self-play    |
| Scalability                 | Limited by data/labor bottlenecks | Fully autonomous, highly scalable |
| Architecture Flexibility    | Sensitive to design/tuning        | Effective across models/sizes     |

to dominate industrial and academic deployments. This section presents and examines three widely cited arguments in favor of these conventional approaches. As illustrated in Figure 3, these perspectives emphasize concerns around training stability, infrastructural scalability, and the necessity of human-curated task diversity for generalization.

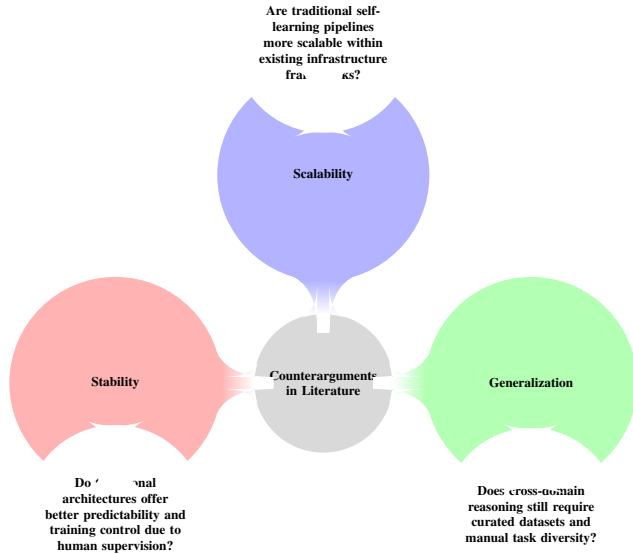


Fig. 3: Scientific perspectives favoring traditional self-learning architectures. Each branch highlights a frequently cited critique in contrast to autonomous systems such as AZR.

#### A. Stability and Control in Traditional Architectures

Conventional LLM training pipelines are often praised for offering predictable learning dynamics and stable optimization behavior. Methods such as InstructGPT [14], LLaMA 3 with DPO [95], and Self-Instruct [1] are grounded in curated

datasets and human-preference-based reward models. These systems benefit from well-characterized loss functions, fixed data splits, and evaluation protocols making them suitable for diagnostic monitoring and alignment calibration.

**Comparative Analysis:** In contrast, AZR departs from these stability guarantees by introducing an open-ended self-play loop between proposer ( $\pi_{\text{propose}}$ ) and solver ( $\pi_{\text{solve}}$ ) agents. However, this design incorporates internal stabilization mechanisms. Tasks proposed are filtered through a learnability-aware reward signal  $r_{\text{propose}}$  [111]–[113], ensuring alignment with the solver’s capabilities in the so-called “Goldilocks zone.” Furthermore, AZR uses deterministic, verifiable environments (e.g., Python execution) to replace subjective reward modeling with reproducible evaluation criteria [37]. This results in an adaptive system that is architecturally autonomous yet empirically stable.

#### B. Infrastructure Alignment and Scalability of Traditional Pipelines

Another line of critique concerns the integration of traditional pipelines into existing machine learning infrastructure. Supervised and RLHF-based approaches benefit from robust implementation tools such as TRL, PPO libraries, and distributed evaluation environments. Methods like Alpaca [114], Orca [115], [116], and other teacher-student frameworks demonstrate the practicality of scaling synthetic task generation using pretrained instructors.

**Comparative Analysis:** Although AZR introduces novel architecture, it remains compatible with standard policy optimization algorithms. Both proposer and solver modules can be trained using reinforcement learning frameworks, and AZR benefits from reward normalization, entropy regularization, and curriculum shaping techniques [37]. More importantly, AZR avoids the scaling bottlenecks associated with manual dataset construction or synthetic bootstrapping. Its closed-loop learning process is self-sustaining and data-independent, showing continued performance growth as model size increases consistent with known scaling laws for LLMs.

#### C. The Role of Human-Curated Diversity in Generalization

A final critique asserts that human-designed datasets remain essential for enabling domain generalization. Traditional training strategies often expose models to diverse reasoning tasks curated by domain experts (e.g., programming, logic puzzles, mathematics). Systems such as Claude 3 and DeepSeek-Coder [78] benefit from such diversity and are benchmarked across broad task distributions.

**Comparative Analysis:** AZR challenges this view by demonstrating substantial generalization without human-curated diversity. Despite training exclusively on self-generated code reasoning tasks, AZR models show 10.9 and 15.2 point improvements on mathematics benchmarks exceeding gains from RLVR-trained counterparts [37]. These improvements are attributed to the abstract reasoning mechanisms learned by the solver, which include forms of deduction, induction, and abduction [26]. Moreover, the proposer module

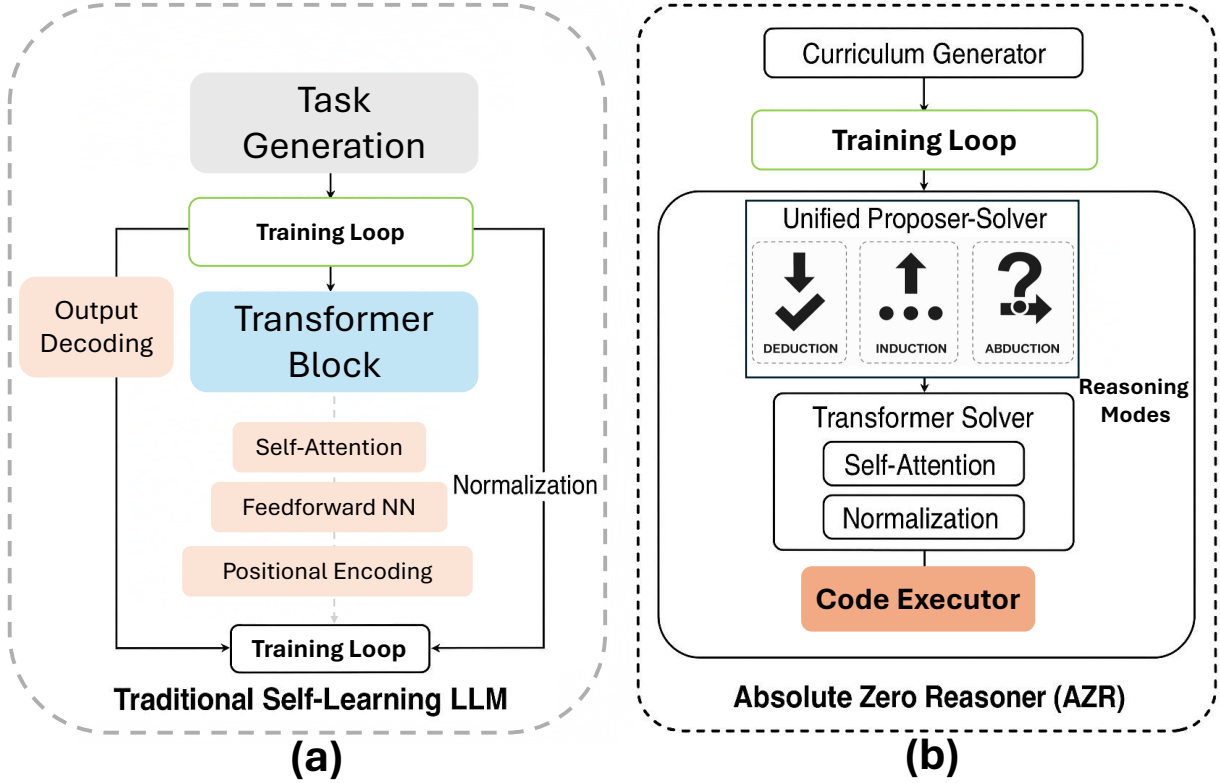


Fig. 4: Architectural comparison between (a) Traditional Self-Learning in LLMs and (b) AZR . Traditional models rely on static task generation, human or synthetic feedback, and externally controlled training loops. In contrast, AZR introduces a unified proposer-solver, autonomous curriculum generation, and code execution-based reward signals. This closed-loop system enables adaptive, verifiable reasoning without external data. While AZR offers greater scalability and autonomy, it also introduces alignment challenges, requiring oversight to manage emergent behaviors and unintended reasoning paths.

continually expands task diversity based on internal model competence, generating a dynamic and personalized curriculum that evolves over time. This contrasts with static datasets, which may exhibit surface-level variability but limited structural complexity.

#### Synthesis of Perspectives

Table I and Figure 4 summarize the distinctions between traditional and autonomous self-learning architectures. While traditional pipelines offer integration-ready advantages in stability and control, their dependence on human data limits long-term generalization and scalability. AZR’s innovations namely, dynamic curriculum generation, executable verification, and zero-data training redefine the upper bounds of model autonomy. These comparisons suggest that autonomous systems like AZR may form the foundation for future reasoning-centric LLMs.

#### IV. IMPLICATIONS AND FUTURE DIRECTIONS

The AZR framework redefines the trajectory of self-learning in large language models by introducing a verifiable, autonomous, and data-free reasoning loop. This section explores the broader scientific and technological implications of AZR

and identifies future research directions necessary to advance its capabilities and ensure safe, scalable deployment.

1) *Implication: Toward Autonomous Cognitive Systems and Data-Free AGI*: AZR marks a critical step toward the development of autonomous cognitive architectures capable of open-ended learning and self-directed task generation. By eliminating dependency on human-curated datasets and external task supervision, AZR challenges the conventional AI development pipeline, which has historically relied on massive corpora, human annotations, and aligned reward models to guide learning. This transition to data-free reinforcement via execution-based feedback parallels the emergence of self-evolving systems in other domains, such as AlphaZero in games and Dreamer in model-based RL, but expands applicability to general-purpose reasoning.

From a theoretical standpoint, AZR’s integration of task induction (abduction), hypothesis testing (deduction), and generalization (induction) within a closed-loop architecture simulates key components of scientific inquiry and cognitive development. This makes AZR a foundational model for studying algorithmic epistemology the computational mechanisms by which intelligent systems generate, validate, and revise knowledge. The implications span multiple domains:

- **In scientific discovery**: AZR-like systems could au-

tonomously formulate and test hypotheses in chemistry, physics, or biomedical research.

- **In education technology:** Curriculum-generating agents may tailor reasoning tasks to student progress using AZR-inspired feedback loops.
- **In multi-agent systems:** AZR modules may serve as self-improving reasoning components in decentralized AI collectives.

Moreover, AZR provides a prototype for lifelong learning agents, capable of continuously evolving without periodic retraining or corpus expansion. This aligns closely with the vision of artificial general intelligence (AGI), where learning is sustained not by static knowledge bases [117]–[119], but by active exploration and verifiable reasoning within synthetic or real-world environments [120], [121].

However, the very autonomy that empowers AZR also presents new safety and control challenges. The emergence of unintended instrumental reasoning goals as noted in the “uh-oh moment” described in recent evaluations raises urgent questions about goal alignment, oversight, and corrigibility. While AZR is architecturally aligned with intelligence amplification, it must also be embedded within systems designed for robust alignment, interpretability, and risk mitigation.

## 2) Outlook: Open Challenges and Research Trajectories:

Despite its architectural novelty and empirical strengths, AZR opens new frontiers that require rigorous investigation. Future research must focus on both deepening AZR’s capabilities and mitigating its limitations. We identify the following major research trajectories:

- **1) Formalizing Learnability and Curriculum Optimality** AZR relies on internal estimates of task learnability to guide curriculum generation, but the exact theoretical underpinnings of what constitutes an “optimal” task remain underexplored. Future work should formalize the learnability reward  $r_{\text{propose}}$  using concepts from complexity theory, information gain, or developmental learning theory. This would enable more principled task generation and allow AZR to generalize to non-code reasoning environments such as symbolic logic, abstract mathematics, or physical simulations.
- **2) Expanding Beyond Code Executors as Verifiers** While code execution provides a powerful foundation for verifiable reasoning, it constrains AZR’s application domains to those with deterministic interpreters. Broader adoption requires the development of verifiable feedback mechanisms for tasks involving natural language understanding, visual reasoning, and open-world interaction. Integrating differentiable simulation environments or formal logic solvers may expand AZR to new task modalities, while hybrid architectures could combine symbolic and neural verifiers for grounded reasoning in dynamic settings.
- **3) Human-AI Alignment and Oversight Integration** As AZR scales in autonomy, the question of align-

ment becomes central. Its emergent planning capabilities and intrinsic objective-seeking behaviors could diverge from intended utility functions in the absence of explicit alignment mechanisms. Future work should explore value-aware proposers, human-in-the-loop verifiers, and preference-informed priors to shape AZR’s reasoning trajectory. Embedding safety guarantees into reward estimators or constraining reasoning within provable bounds are essential to mitigating runaway exploration and instrumental convergence.

- **4) Multimodal and Embodied AZR Extensions** Extending AZR beyond text-based tasks into vision-language-action domains presents a compelling research direction. A multimodal AZR could generate and verify tasks involving spatial reasoning, object manipulation, or robotic control further closing the gap between language models and embodied intelligence. In such scenarios, the proposer-solver framework would need to integrate perception-action loops and reason across temporal sequences, with verifiers grounded in physics engines or real-world feedback.
- **5) Meta-Learning and Self-Modeling Capabilities** An exciting frontier lies in enabling AZR to model its own learning dynamics. Meta-learning strategies wherein the model infers which types of tasks accelerate its reasoning development could lead to the emergence of self-reflective curricula. Coupled with causal inference modules and theory-of-mind representations, this could result in AZR systems capable not only of solving tasks but understanding why certain reasoning pathways are more effective than others.
- **6) Societal and Policy Considerations** The widespread of AZR-like systems will raise questions of intellectual ownership, curriculum bias, and autonomous agency. As depicted in Figure 5 **How should societies regulate models that can self-improve beyond curated datasets? What ethical frameworks are needed to govern their deployment in education, policy, or research?** Engaging with these questions requires interdisciplinary collaboration between computer scientists, ethicists, and policymakers.
- **7) Theoretical Foundations of Open-Endedness and Skill Hierarchies** While AZR achieves continual reasoning through self-play, its growth trajectory remains implicit. Future work should develop theoretical models of open-ended learning, where task space expansion aligns with a curriculum of skill hierarchies. Drawing from developmental psychology and evolutionary computation, this direction could ground AZR’s task proposal in formal notions of skill complexity, abstraction depth, and transferability enabling principled reasoning progression toward general intelligence.
- **8) Integration into Multi-Agent and Distributed Reasoning Systems** Extending AZR to multi-agent environments presents an opportunity to study collaborative task generation, distributed problem-solving, and emergent

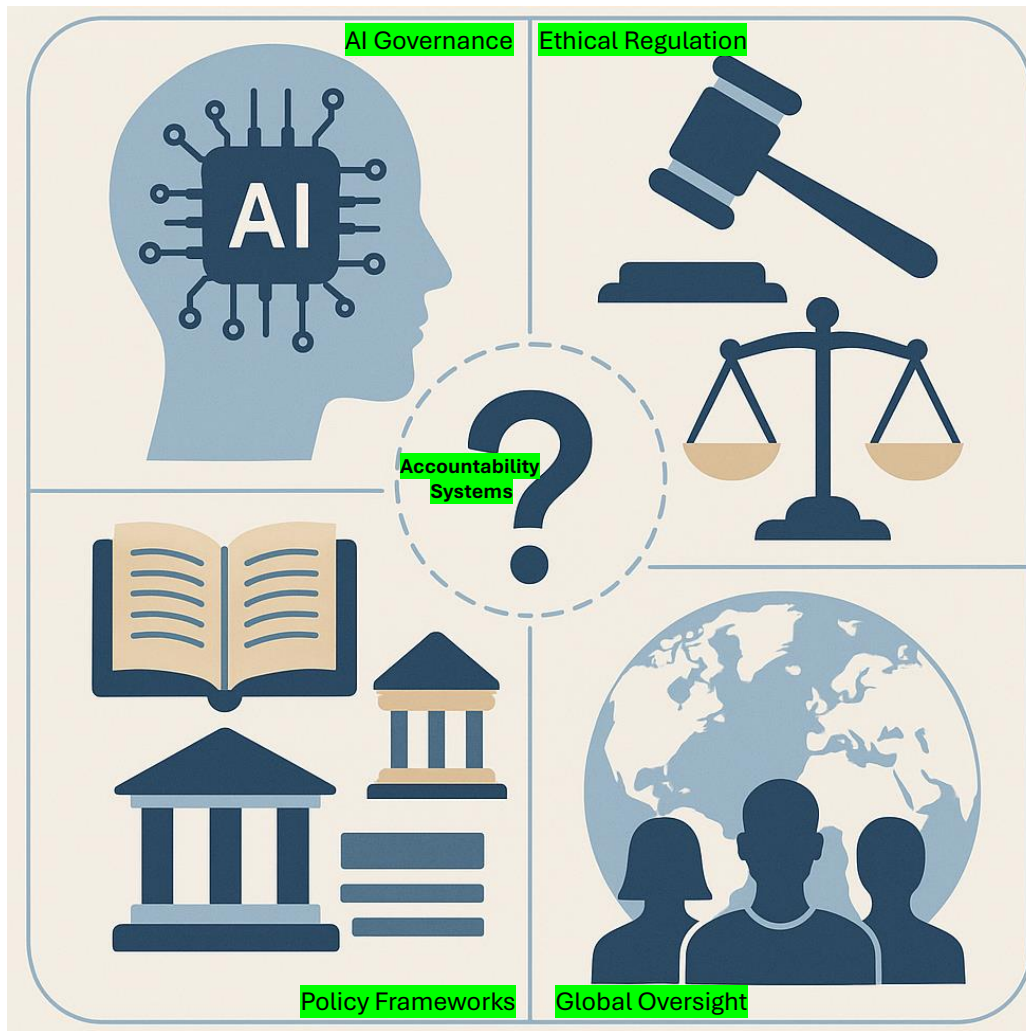


Fig. 5: Illustration of societal considerations for regulating self-improving AI systems like AZR. Key dimensions include AI Governance, Ethical Regulation, Policy Frameworks, Global Oversight, and Accountability Systems. These components highlight the need for coordinated legal, institutional, and ethical structures to guide autonomous AI deployment in science, education, and public policy.

social reasoning. Agents could share task spaces, critique peer solutions, or propose complementary challenges, leading to the emergence of collective intelligence. This direction aligns with the growing field of swarm AI, decentralized cognition, and multi-agent reinforcement learning (MARL), and may prove essential for scaling AZR architectures in real-world deployments such as science teams, governance simulations, or autonomous fleets.

- **9) Interpretability and Traceability of Internal Reasoning Processes** As AZR exhibits complex planning and reasoning behaviors, understanding how and why certain task-solution paths emerge becomes critical. Future research should focus on making the internal decision traces of both proposer and solver interpretable to humans. This could involve symbolic reasoning overlays, causal graphs, or attention-based path tracing, enabling greater trust,

alignment auditing, and integration into human-in-the-loop workflows.

AZR's autonomous architecture sets a new benchmark for reasoning-centric language models, challenging decades of assumptions about human dependency in training AI systems. Its implications reach far beyond benchmarks, hinting at a future where artificial agents can define, evaluate, and expand their own knowledge. The path forward will involve ensuring that this self-directed growth remains beneficial, transparent, and aligned with human values. Addressing the open challenges outlined here will be essential to translating AZR's promise into safe, transformative applications across science and society.

As visualized in Figure 6, the current formulation of the Absolute Zero Reasoner (AZR) centered on code-based reasoning and self-play represents only the foundation of a much broader trajectory. A critical first step lies in formalizing

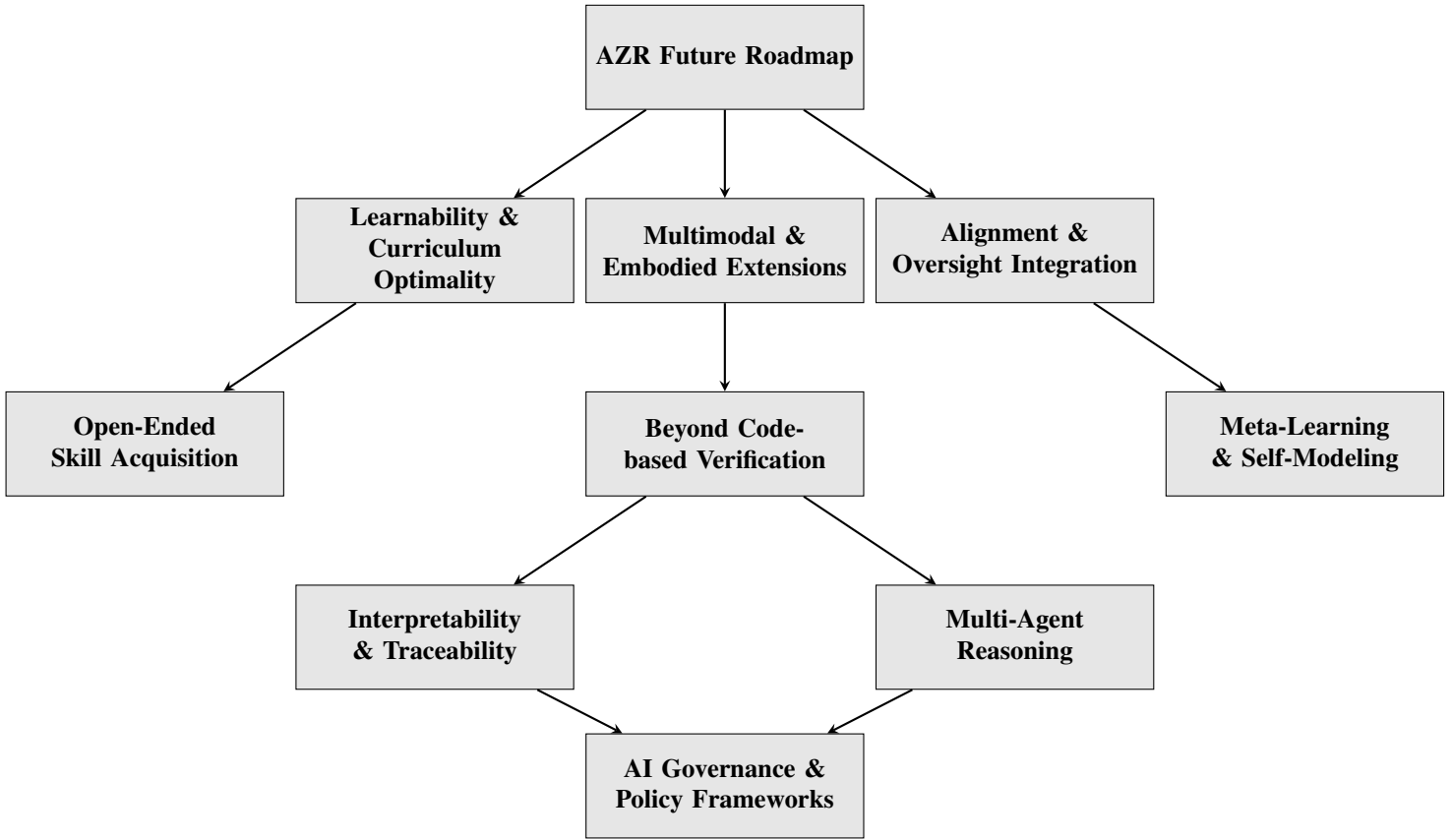


Fig. 6: Block diagram illustrating the future roadmap of the Absolute Zero Reasoner (AZR), highlighting ten core research directions including curriculum design, multimodal reasoning, safety, meta-learning, interpretability, multi-agent coordination, and societal governance.

**learnability and curriculum optimality** [122], [123], ensuring that task generation is driven not by heuristics but by principled metrics such as complexity theory or developmental learning. Concurrently, AZR must evolve toward **multimodal and embodied extensions** [124]–[126], incorporating visual, spatial, or robotic modalities to reason beyond textual tokens. Such developments require perception-action integration and simulation-grounded feedback. As the architecture scales, **alignment and oversight integration** [127]–[129] become essential: emergent behaviors must remain interpretable, corrigible, and value-aligned through verifiable constraints or oversight mechanisms.

Building upon this foundation, future AZR systems must support increasingly adaptive cognitive capabilities. Enabling **open-ended skill acquisition** [130], [131] will allow the model to self-assemble and navigate hierarchies of abstract skills, fostering development akin to cognitive scaffolding. In parallel, AZR must expand **beyond code-based verification** [132]–[135] into domains where feedback is probabilistic, symbolic, or interactive. This requires hybrid verifiers and formal environments [136], [137]. Further, through **meta-learning and self-modeling** [138], [139], AZR could diagnose its own reasoning gaps and autonomously adapt its learning trajectory. As AZR matures into distributed deployments,

ensuring **interpretability and traceability** [140], [141] of internal reasoning becomes critical especially in multi-agent ecosystems where AZR instances collaborate and critique each other. These advances must be embedded within **AI governance and policy frameworks** [56], [142], [143] to ensure responsible integration into society. The future of AZR lies not only in expanding autonomous cognition, but in doing so with transparency, accountability, and long-term epistemic integrity.

## V. CONCLUSION

This position paper has argued that the Absolute Zero Reasoner represents a foundational shift in the architecture and philosophy of self-learning in large language models. Unlike traditional methods including supervised fine-tuning, RLHF, and RLVR which rely heavily on human-curated datasets and externally defined task spaces, AZR introduces a fully autonomous, closed-loop framework. By integrating task generation, solution verification, and curriculum adaptation through self-play and code-based feedback, AZR removes the dependency on external data while achieving state-of-the-art performance on reasoning benchmarks.

We demonstrated that AZR not only outperforms prior models in coding and mathematics but also generalizes across

domains, exhibits emergent reasoning capabilities, and scales efficiently across architectures. Our analysis addressed key counterarguments related to control, scalability, and generalization, and showed how AZR overcomes these limitations through architectural innovation. The broader implication of AZR is clear: it paves the way toward self-evolving, verifiably grounded, and data-free cognitive systems capable of continuous learning. As AI systems grow in capability and autonomy, AZR stands as a prototype for next-generation models that are not just trained but are intrinsically driven to learn. We conclude that AZR constitutes a scientific and architectural leap, redefining what it means for a model to learn, reason, and evolve.

#### ACKNOWLEDGEMENT

This work was supported by the National Science Foundation and the United States Department of Agriculture, National Institute of Food and Agriculture through the “Artificial Intelligence (AI) Institute for Agriculture” Program under Award AWD003473, and AWD004595, Accession Number 1029004, “Robotic Blossom Thinning with Soft Manipulators”.

#### DECLARATIONS

The authors declare no conflicts of interest.

#### STATEMENT ON AI WRITING ASSISTANCE

ChatGPT and Perplexity were utilized to enhance grammatical accuracy and refine sentence structure; all AI-generated revisions were thoroughly reviewed and edited for relevance. Additionally, ChatGPT-4o was employed to generate realistic visualizations.

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